Exam 2: Exploring Data – Version C

Exam Agreement

This exam is an **individual assessment** and you should neither offer help to nor receive help from another person.

Cell phones are to be handed in to the instructor and all other electronic devices must be stored in bags at the front of the room.

No smart glasses or other wearable electronic devices are to be powered on during this exam except with accommodations.

You may use **no physical or electronic** resource to complete the work.

By signing, you agree that you have neither given nor received unauthorized aid on this examination.

Printed Name: _____

Chapman ID: _____

Signed: _____

Section: _____ **Date:** _____

Include this signed page as the first page of your submitted work.

Exam Instructions

You must answer every question on the exam to the best of your ability. Include all appropriate code, syntax, and functions that would lead to the code to run successfully.

Question 1: Loading Data

Your data file "rhabadophis-data.csv" contains your measurements of a population survey of Sri Lanka keelback snakes (*Rhabdophis ceylonensis*), which are endangered. The measurements contain total length (snout-to-vent) of each snake captured and measured and the date the snake was captured and measured. The first two lines contain information about the collection site and other notes, and the third line has column names. Write code that will read this CSV into R.

Question 2: Cleaning Data

Originally saved from Excel, your data set from Question 1 now contains a lot of empty rows and columns. There are also some missing observations, where the initial value was removed. Using the tidy pipe operator, clean the data for three things: standardize the column names, remove the empty rows and columns, and remove every observation that is incomplete.

Question 3: Basic Data Analysis

Last year, the Sri Lankan government started releasing juvenile keelbacks of a certain size into your study area to try and support the endangered population. They want to see if your data on snake length – which directly correlated with the age of the snake – will reveal whether this reintroduction program is leading to more snakes after only one year. What plot would you use to answer their question and why? Write the code for the plot using either base graphics or the `ggplot2` package.

Question 4: Troubleshooting Compound Code

You are attempting to load the robo-mussel data set with the following code:

```
robo_dat <- read.table("robo-mussel.csv")
```

But you get the error:

```
Error in scan(file = file, what = what, sep = sep, quote = quote, dec = dec,  :
  line 2 did not have 4 elements
```

What does this error mean, and how do you fix it?

Question 5: Plotting with Base Graphics

Using base graphics, create two box plots of the `ToothGrowth` data set (Fig. 1), each one showing length versus dose for one of the two supplement (`VC` and `OJ`).

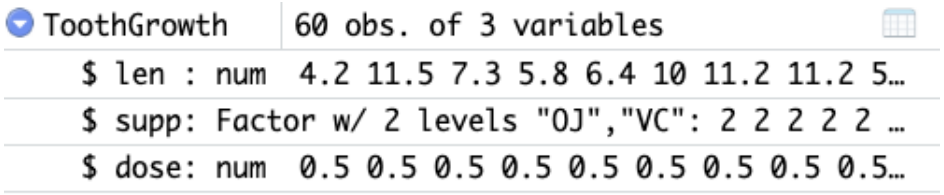


Figure 1: The ToothGrowth data set.

Question 6: Plotting with ggplot2

Using `ggplot2` and the `ToothGrowth` data set (Fig. 1), create one box plot showing length versus dose for both supplements as filled-color boxes.

Question 7: How We See

"Mach banding" or the Chevreul illusion causes the brain to misinterpret the shades of colors where they make contact with other shades of color. Describe at the cellular level how this illusion works.

Question 8: Visual Processing and Attention

Visual working memory contains three to five slots that can be filled with a variety of different things to help us understand the visual world around us. Describe the two sources that fill these spots and their role in the process of visual processing.

List of Functions for Exam 2

- `+, -, *, /, ^` – arithmetic operators
- `sqrt()` – square root
- `log()` – logarithm
- `seq()` – sequence creation
- `rep()` – value/vector repetition
- `sort()` – vector sort
- `length()` – determine vector length
- `[ ]` – vector subset/extraction, list slicing, multiple members
- `sum()` – sum all vector elements
- `prod()` – multiply all vector elements
- `mean()` – averages all vector elements
- `sd()` – takes standard deviation of all vector elements
- `matrix()` – create a matrix
- `rbind()` – create a matrix (bind rows)
- `cbind()` – create a matrix (bind columns)
- `dim()` – get matrix dimensions
- `nrow()` – get number of rows
- `ncol()` – get number of columns
- `[ , ]` – matrix/array/data frame subsetting
- `diag()` – diagonal elements/identity matrix
- `array()` – create an array
- `list()` – create a list
- `[[ ]]` – unnamed member reference list
- `$` – get named member/variable
- `data.frame()` – create data frames
- `==, !=, >, <, >=, <=` – relational operators
- `any()` – checks whether any entries are true
- `all()` – checks whether all entries are true
- `&&, &, ||, |, !` – logical operators
- `which()` – determines indices of `TRUE`s
- `nchar()` – gets the number of characters in a string
- `cat()` – concatenates strings (no return)
- `paste()` – pastes strings (returns a string)
- `\` – escape character
- `substr()` – Subsets a string
- `sub()`, `gsub()` – string matching and replacement
- `factor()` – creates a factor vector
- `levels()` – gets levels of a factor
- `cut()` – creates factor from continuum
- `is.infinite()` – element-wise check for `Inf`
- `is.finite()` – element-wise check for finiteness
- `NaN` – value for invalid numerics
- `is.nan()` – element-wise check for `NaN`
- `NA` – value for missing observation
- `is.na()` – element-wise check for `NA` or `NaN`
- `na.omit()` – delete all `NA`s and `NaN`s
- `NULL` – value for empty
- `is.null()` – check for `NULL`
- `attributes()` – list explicit attributes
- `attr()` – obtain specific attribute
- `dimnames()` – get array dimension names
- `class()` – get object class
- `is._()` – object-checking functions
- `as._()` – object-coercion functions
- `unlist()` – object-coercion from list to vector
- `runif()` – draws pseudo-random numbers from a uniform distribution
- `sample()` – draws numbers from a sample given by user

- `plot()` – makes a plot in base graphics
- `~` – creates formula object
- `points()` – adds points to a plot in base graphics
- `lines()` – adds points to a plot in base graphics
- `legend()` – adds a legend to a plot in base graphics
- `pivot_wider()` – restructure data to wide format (tidyr)
- `pivot_longer()` – restructure data to long format (tidyr)
- `ggplot()` – initiates a plot using ggplot2
- `geom_x()` – geometric object layers in ggplot2
- `ggtitle()` – title a ggplot
- `labs(), xlab(), ylab()` – alter axis labels in ggplot2
- `lims(), xlim(), ylim()` – alter axis limits in ggplot2
- `summary()` – produces summary stats on a data frame
- `head()` – prints first lines of a data frame
- `tail()` – prints last lines of a data frame
- `IQR()` – computes interquartile ranges
- `median()` – computes median
- `table()` – builds a contingency table of counts
- `quantile()` – produces sample quantiles
- `tapply()` – apply function to groupings in data set
- `apply()` – apply function to row or column of data set
- `read.table()` – imports generic text file
- `read.csv()` – imports comma separated value file
- `clean_names()` – standardizes column names in data frame (janitor)
- `remove_empty()` – removes empty columns and/or rows in data frame (janitor)
- `%>%` – tidy pipe operator (tidyr/janitor)
- `write.csv()` – exports object to text csv file